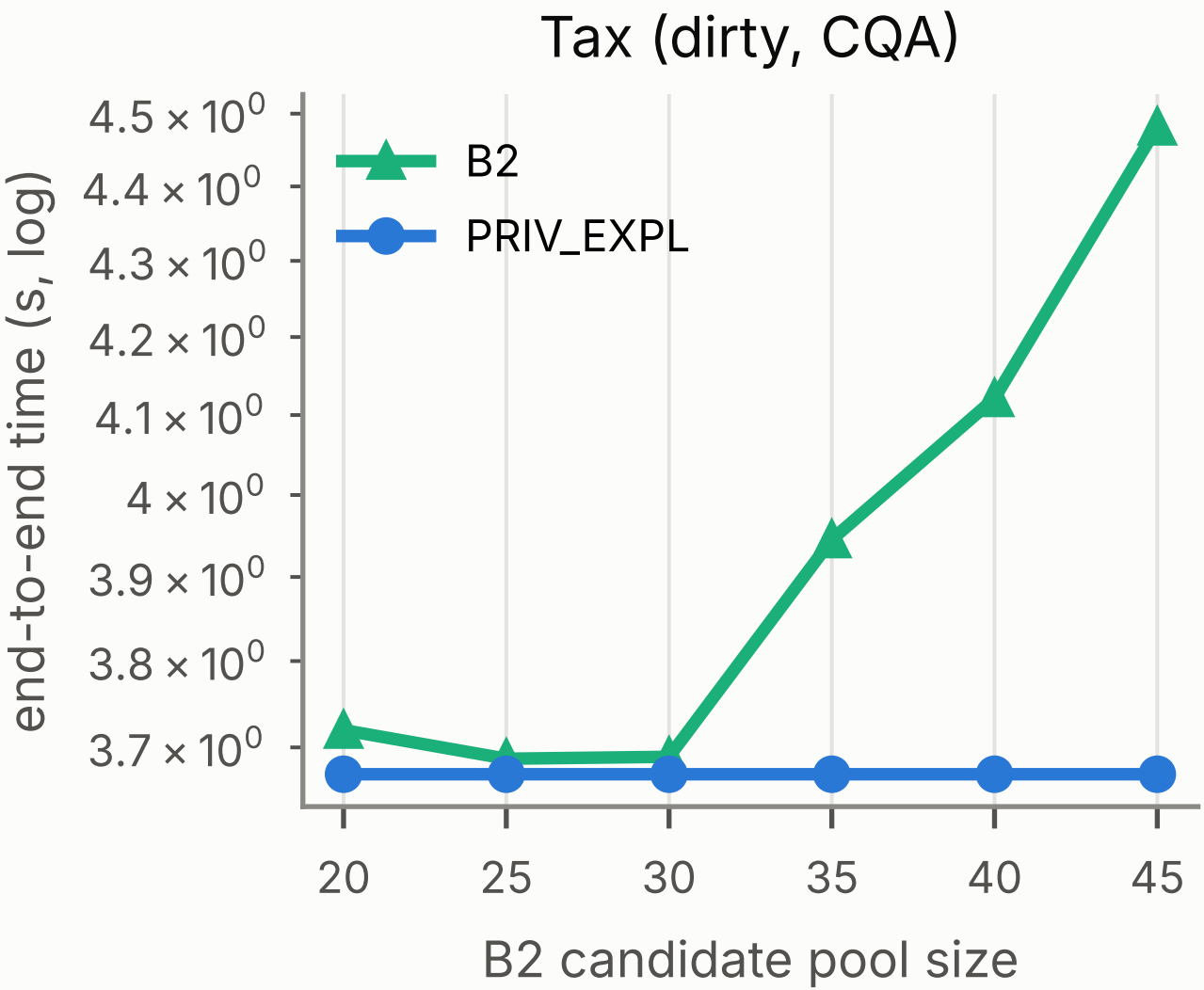B2 grows combinatorially with its candidate pool (end to end, k = 5)